Statistics 6910, Section 003, Fall 2013, Quiz 3 (26 Points)

Wednesday, September 25, 2013

Your Name: ____________________

Question 1: Matrices and Apply (**26 Points**)

The matrix `mat` is created as follows in R:

```
> mat = matrix(c(20, 10, 40, 10, 0, 30, 15, 45, 0, 30), nrow = 5)
```

The content of `mat` is:

```
> mat

     [,1] [,2]
[1,]   20   30
[2,]   10   15
[3,]   40   45
[4,]   10    0
[5,]    0   30
```

This matrix may represent a common two–way table where each element of the matrix lists the counts of observations for each combination of the factors of two categorical variables.
Answer the following questions. Note that all R expressions are correct, i.e., there will be no error messages! If you are uncertain, make your best (educated ☺) guess!

1. The matrix `mat` has _______ columns and _______ rows.
2. The R expression `mat[2, -2]` returns:

3. The R expression `mat[c(5, 3), c(FALSE, TRUE)]` returns:

4. The R expression `mat >= 30` returns:

5. What is the grand (overall) total for this table? Indicate the simplest R expression and the result of your expression.

6. To calculate conditional probabilities, we need the totals for each row (and column). Here, just indicate the R expression for the row totals and the result of your expression.

7. What does the following R expression do? Indicate the numerical result of this expression and provide a statistical explanation (yes, this does exactly what we want it to do!):

```
sapply(1:2, function(x) mat[, x] / sum(mat[, x])) * 100
```

For the next **six** R expressions, just indicate the result. Look carefully whether `mat` changes or not. If it changes, continue with the modified content of `mat` in the next question. Recall the content of `mat`:

```
> mat

     [,1] [,2]
[1,]   20   30
[2,]   10   15
[3,]   40   45
[4,]   10    0
[5,]    0   30
```

8. `mat + 1`

9. `mat^2`

10. `mat / mat`

11. `mat + c(5, 10)`

12. `mat[apply(mat, 1, sum) >= 50, ] = -999; mat`

13. `mat[] = 0; mat`